Quantitative Discovery of Qualitative Information: A General Purpose Document Clustering Methodology

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- (We focus on texts, our methods apply more broadly)

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- Now imagine choosing the *optimal* classification scheme by hand!
- Its no surprise that automated algorithms can help, but which algorithms?

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- Deep problem in cluster analysis literature: no way to know which method will work ex ante

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  - An organized list will make the search possible

Our Idea: Meaning Through Geography

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Call....Pembroke TelNo-617 436-5353
Carter J Veal Co
48 Newmarket Sq Rox 02118....617 442-1775
Carter James
1573 Cambridge St Cam 02138....617 492-1214
James 182 Fisher Av Roxbury 02120....617 739-2193
James
37 Gold Star Rd Cambridge 02140....617 876-8841
Jas L 14 Roseberry Rd Mat 02126....617 361-0773
Jane 114 Adena Rd Newton 02465....617 964-0435
Jeffrey 41 Warren Av Bos 02116....617 426-5994
John 11 Mansfield Bri 02134....617 987-2163
John 327 Summer Bos 02210....617 423-4334
John 40 Westwind Rd Dor 02125....617 282-1235
June O 329 A Summit Av Bri 02135....617 734-6109
K 38 Browning Av Dorchester 02124....617 265-8456
K 17 Esmond Dorchester 02121....617 282-1593

Carter Nella E
333 Maschsts Av Bos 02115....617 267-6483
Nicholas S F
115 Randolph Av Mil 02186....617 698-5307
Nick 21 Fairfield Bos 02116....617 267-5222
Nick & Debbi
196 Herrick Rd Newton 02459....617 527-0480
Nicole....617 698-0713
Norman G
38 Chickatawbut Dor 02122....617 822-1203
P 94 Crestwood Pk Rox 02121....617 427-4754
P E 501 E Sixth S Bos 02127....617 268-4213
P L 44 Hutchings Rox 02121....617 427-9170
P R 91 Bynner Jam 02130....617 983-8692
Paul & Constance
114 Anawan Av W Rox 02132....617 325-2036
Paul E 501 E Sixth St S Bos 02127....617 268-4546
Paul M 27 Union Bri 02135....617 787-2115
Carter Pile Driving Inc 17 Beaver Ct
Framingham 01702.... Wellesley TelNo-781 235-8488
Carter Prudence
46 Franklin Watertown 02172....617 393-3782
Prudence
46 Franklin Watertown 02172....617 926-7063
Reginald
106 Brunswick Dorchester 02121....617 541-2843
Renee & Andrew
10 Walnut Bos 02108....617 720-3765
Carter Rice Dowd
Bulkley Dunton Publishing 163 Main Wilmington 01887
Toll Free-Dial '1' & Then....800 638-1671
Cust Svc-Industrial Prod 613 Main Wilmington
Toll Free-Dial '1' & Then....800 619-7447
Cust Svc-Printing 613 Main Wilmington
Toll Free-Dial '1' & Then....800 648-7447
Headquarters 613 Main Wilmington 01887
Call....978 988-7447
Ingalls Cronin 163 Main Wilmington 01887
Toll Free-Dial '1' & Then....800 638-1673
Carter Richard
1079 Commwlth Av Brighton 02215....617 987-0836
Richard A 97 Mt Vernon Bos 02108....617 566-7293
Carter Richard A MD
170 Commwlth Av Bos 02116....617 267-0710
Carter Richard K
15 Mercer S Bos 02127....617 268-0448
Robert L 175 Richdale Av Cam 02140....617 864-1535
Roger 150 St Botolph Bos 02115....617 424-6148
Roy 44 Concord Av Cam 02138....617 491-6115
Royce 18 Seminary Cha 02129....617 241-0418

$\rightsquigarrow$

$\rightsquigarrow$ We develop a (conceptual) geography of clusterings

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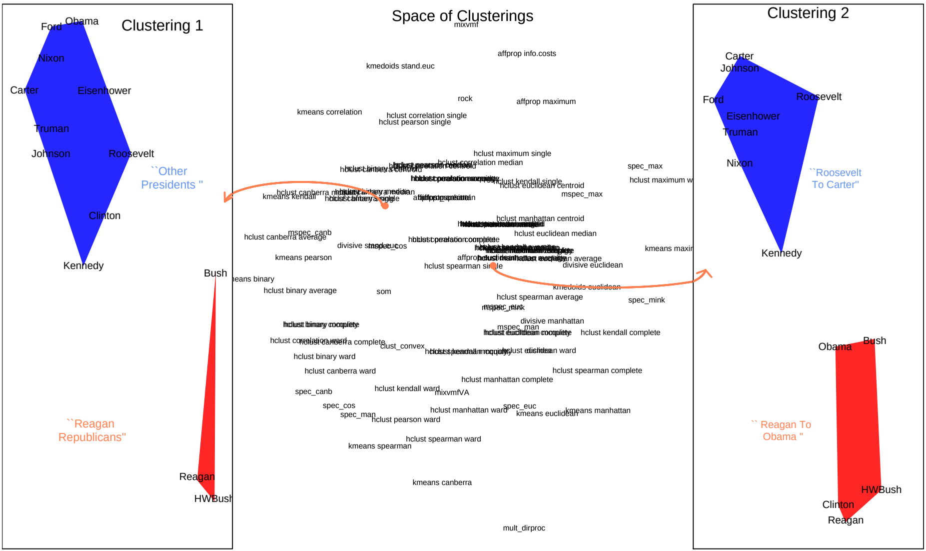
7. $\rightsquigarrow$ Millions of clusterings, easily comprehended (takes about 10-15 minutes to choose a clustering with insight)

Many Thousands of Clusterings, Sorted & Organized

You choose one (or more), based on insight, discovery, useful information, . . .



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- Meila (2007): derives same metric using different axioms (lattice theory)

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  - Discovery $\Rightarrow$ You're the judge

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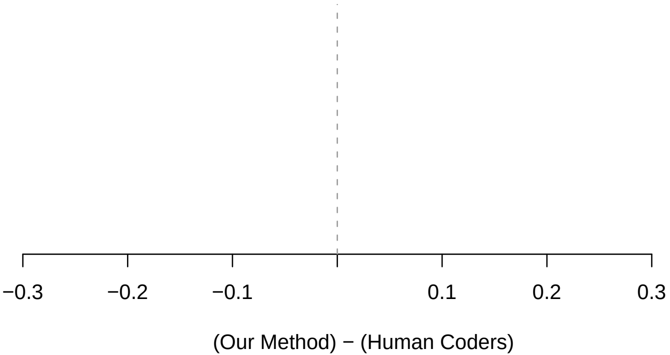

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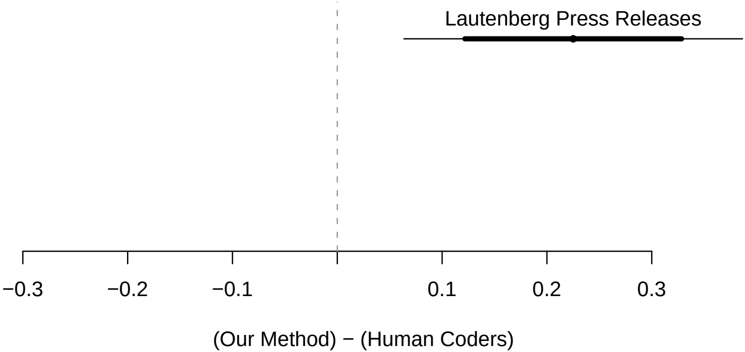

Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, . . .)

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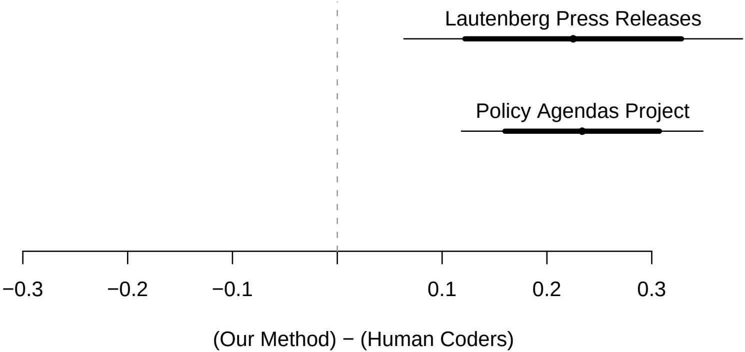

Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, . . .)

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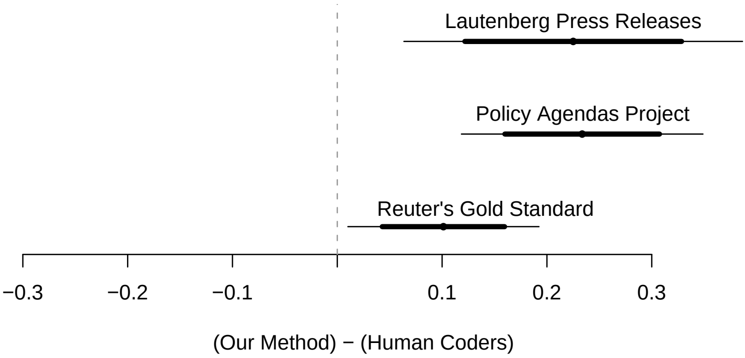

Reuter's: financial news (trade, earnings, copper, gold, coffee, . . .); "gold standard" for supervised learning studies

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"Genetic testing":

Our Method 1 $\rightarrow$ {Our Method 2, K-Means 1, K-means 2} $\rightarrow$ Dir Proc. 1 $\rightarrow$ Dir Proc. 2

Evaluation 3: What Do Members of Congress Do?

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- Apply our method

Example Discovery

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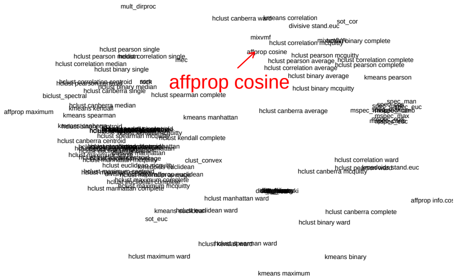

Red point: a clustering by Affinity Propagation-Cosine (Dueck and Frey 2007)

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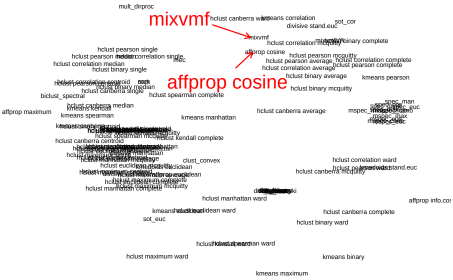

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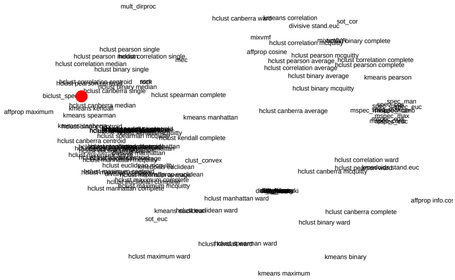
Close to:

Mixture of von Mises-Fisher
distributions (Banerjee et. al.
2005)

Example Discovery

Space between methods:

Example Discovery



Space between methods:

Example Discovery

Space between methods:

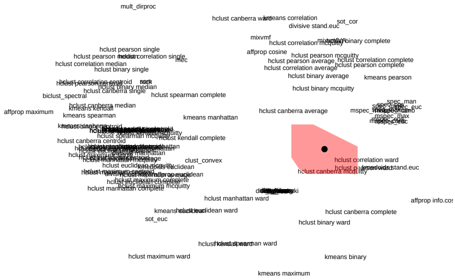
local cluster ensemble

Example Discovery

Example Discovery

Found a region with particularly insightful clusterings

Example Discovery



Mixture:

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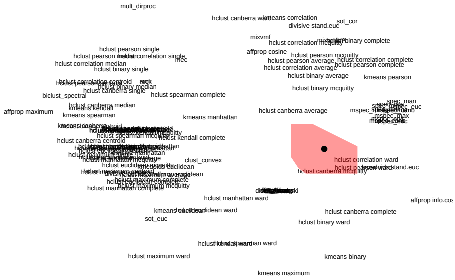
Example Discovery



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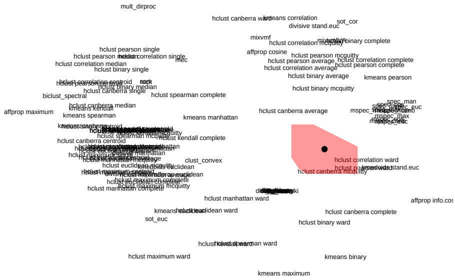

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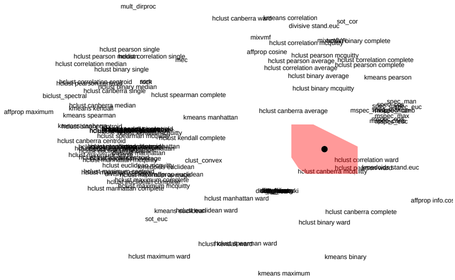

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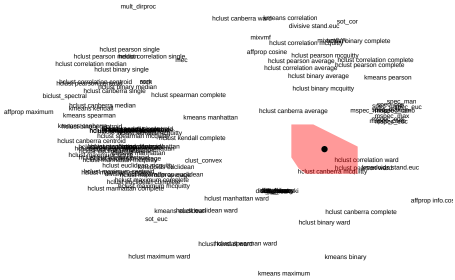

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- 0.05 Kmediods-Cosine
- 0.04 Spectral clustering Symmetric (Metrics 1-6)

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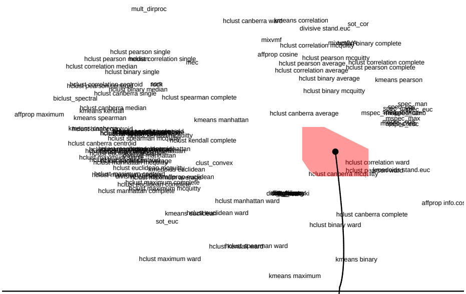
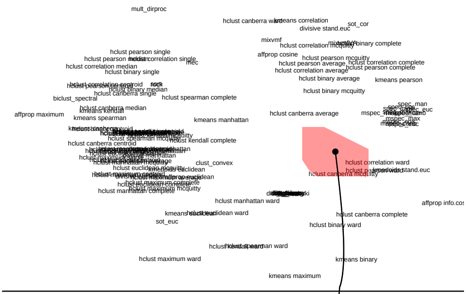


Clusters in this Clustering

Mayhew

Example Discovery



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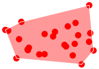


Credit Claiming
Pork

Credit Claiming, Pork:

"Sens. Frank R. Lautenberg (D-NJ) and Robert Menendez (D-NJ) announced that the U.S. Department of Commerce has awarded a $100,000 grant to the South Jersey Economic Development District"

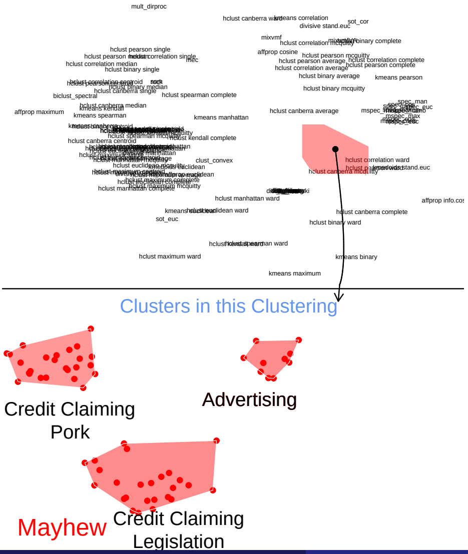
Mayhew

Example Discovery

Credit Claiming, Legislation:

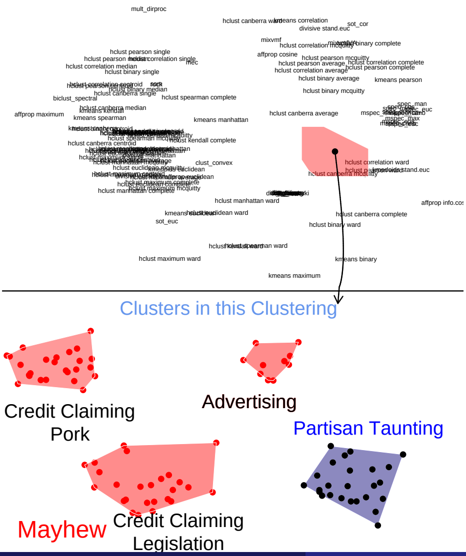
"As the Senate begins its recess, Senator Frank Lautenberg today pointed to a string of victories in Congress on his legislative agenda during this work period"

Example Discovery



Advertising:
"Senate Adopts Lautenberg/Menendez Resolution Honoring Spelling Bee Champion from New Jersey"

Example Discovery: Partisan Taunting



Partisan Taunting:
"Republicans Selling Out Nation on Chemical Plant Security"

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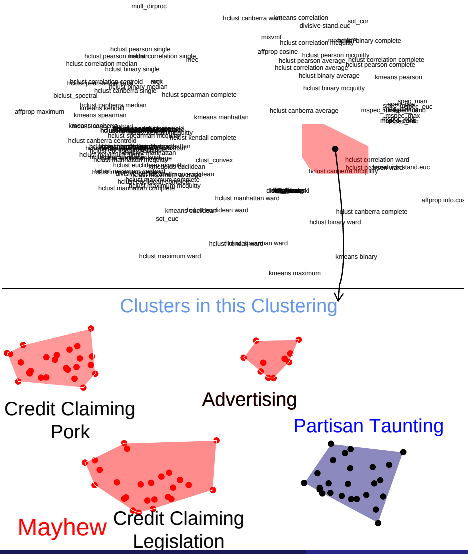

Partisan Taunting: "Senator Lautenberg's amendment would change the name of …the Republican bill…to 'More Tax Breaks for the Rich and More Debt for Our Grandchildren Deficit Expansion Reconciliation Act of 2006' "

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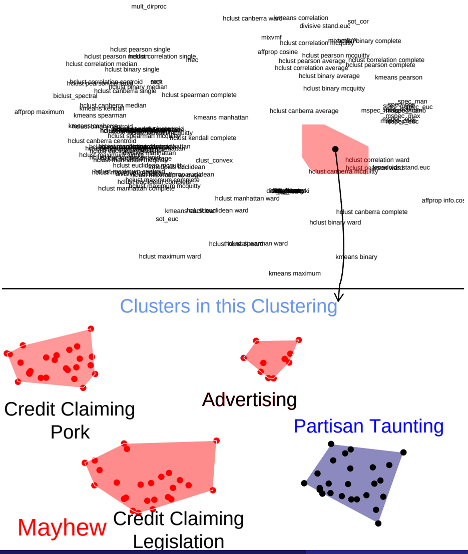
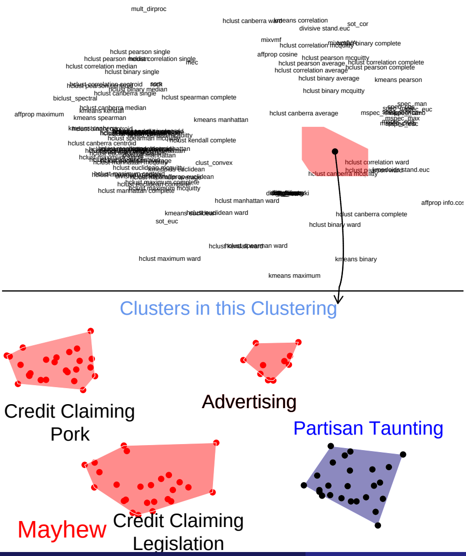


Definition: Explicit, public, and negative attacks on another political party or its members

Example Discovery: Partisan Taunting



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Sen. Lautenberg on Senate Floor 4/29/04

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In Sample Illustration of Partisan Taunting

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- "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then" [Healthcare]
- "Every day the House Republicans dragged this out was a day that made our communities less safe." [Homeland Security]

Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.

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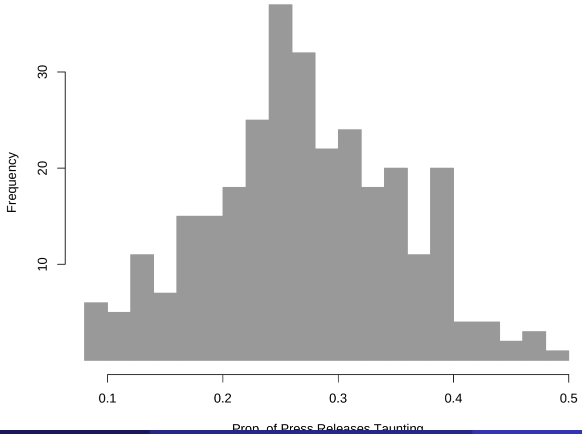

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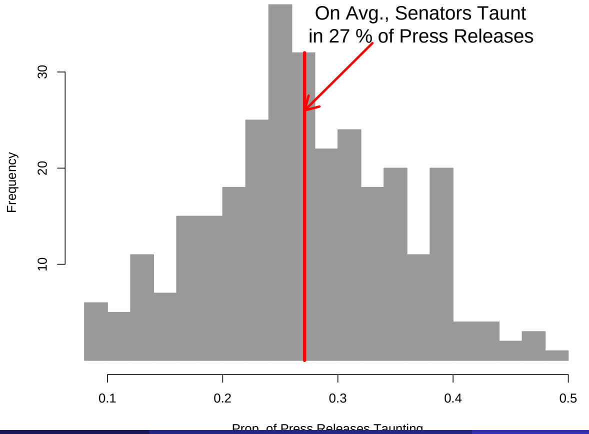

Advancing the Objective of Discovery

1) Conceptualization

Qualitative Methods (reading!)

2) Measurement

Quantitative Methods

3) Validation

Quantitative methods for conceptualization: aiding discovery

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Quantitative methods for conceptualization: aiding discovery

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- Evaluation methods measure progress in discovery

For more information

http://GKing.Harvard.edu